**Math 310** **Fall 2026**
**Homework 6** **Drew Armstrong**

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**1. Various Kinds of First and Second Derivatives in $\mathbb{R}^3$.** For any scalar field $f(x,y,z)$ we define a vector field $\text{Grad}(f)$ and a scalar field $\text{Laplacian}(f)$ by

$$\text{Grad}(f) = \text{“}\nabla f\text{”} = \langle f_x, f_y, f_z\rangle,$$
$$\text{Laplacian}(f) = \text{“}\nabla^2 f\text{”} = f_{xx} + f_{yy} + f_{zz}.$$

and for any vector field $\mathbf{F}(x,y,z) = \langle P(x,y,z), Q(x,y,z), R(x,y,z)\rangle$ we define a vector field $\text{Curl}(\mathbf{F})$ and a scalar field $\text{div}(\mathbf{F})$ by

$$\text{Curl}(\mathbf{F}) = \text{“}\nabla \times \mathbf{F}\text{”} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y\rangle,$$
$$\text{Div}(\mathbf{F}) = \text{“}\nabla \bullet \mathbf{F}\text{”} = P_x + Q_y + R_z.$$

(a) For any scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ check that $\text{Curl}(\text{Grad}(f)) = \langle 0,0,0\rangle$.
(b) For any vector field $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ check that $\text{Div}(\text{Curl}(\mathbf{F})) = 0$.
(c) For any scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ check that $\text{Div}(\text{Grad}(f)) = \text{Laplacian}(f)$.

**2. Conservative Vector Fields.** Consider the vector field

$$\mathbf{F}(x,y,z) = \langle 2x + y, x + z, y\rangle.$$

(a) Check that the curl is zero: $\nabla \times \mathbf{F}(x,y,z) = \langle 0,0,0\rangle$.
(b) It follows from (a) that there exists a scalar field $f(x,y,z)$ satisfying $\nabla f(x,y,z) = \mathbf{F}(x,y,z)$. Find one example of such a field. [Hint: Integrate $\mathbf{F}$ along an arbitrary path starting at some arbitrary point and ending at the point $(x,y,z)$. For the purpose of this calculation let $x,y,z$ be constant.]

**3. Green's Theorem on a Rectangle.** Consider the vector field $\mathbf{F}(x,y) = \langle y^2, x^2\rangle$.

(a) Compute the scalar curl of $\mathbf{F}$.
(b) Integrate the scalar curl of $\mathbf{F}$ over the rectangle with $0 \le x \le 2$ and $0 \le y \le 1$.
(c) Let $C_1, C_2, C_3, C_4$ be the four sides of the rectangle, oriented counterclockwise. Integrate $\mathbf{F}$ along each of these curves and add the results. Check that your answers to (b) and (c) are the same. [Hint: You can parametrize the four sides by

$$\mathbf{r}_1(t) = (0,0) + t(2,0),$$
$$\mathbf{r}_2(t) = (2,0) + t(0,1),$$
$$\mathbf{r}_3(t) = (2,1) + t(-2,0),$$
$$\mathbf{r}_4(t) = (0,1) + t(0,-1),$$

each with $0 \le t \le 1$.]

**4. Current in a Wire.** Imagine a wire of radius $a$ centered around the $z$-axis, with constant electric current traveling in the positive $z$ direction. The magnetic field due to this current is[1]

$$\mathbf{B}(x,y,z) = \begin{cases} \frac{1}{a^2}\langle -y, x, 0\rangle & x^2 + y^2 \le a^2, \\ \frac{1}{x^2+y^2}\langle -y, x, 0\rangle & x^2 + y^2 \ge a^2. \end{cases}$$

Compute the electric current density $\mathbf{J} := \nabla \times \mathbf{B}$. What happens at the boundary $x^2 + y^2 = a^2$?

[1] We ignore all physical constants.

**5. Stokes' Theorem on a Pringle.** Consider the vector field $\mathbf{F}(x, y, z) = \langle -y, x, 1\rangle$ and the pringle-shaped surface $D$ defined by

$$\mathbf{r}(u, v) = \langle u\cos v, u\sin v, u^2\cos v\sin v\rangle,$$

with $0 \le u \le 1$ and $0 \le v \le 2\pi$.

(a) Compute the curl $\nabla \times \mathbf{F}(x, y, z)$.

(b) Compute the flux of the curl $\nabla \times \mathbf{F}$ across the pringle:

$$\begin{aligned}\iint_D (\nabla \times \mathbf{F}) \bullet \mathbf{N}\, dA &= \iint (\nabla \times \mathbf{F})(\mathbf{r}(u, v)) \bullet \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \|\mathbf{r}_u \times \mathbf{r}_v\|\, dudv \\ &= \iint (\nabla \times \mathbf{F})(\mathbf{r}(u, v)) \bullet (\mathbf{r}_u \times \mathbf{r}_v)\, dudv.\end{aligned}$$

[Hint: From the previous homework we have $\mathbf{r}_u \times \mathbf{r}_v = \langle -u^2\sin v, -u^2\cos v, u\rangle$.]

(c) Let $C = \partial D$ be the boundary curve of the pringle. Compute the circulation of $\mathbf{F}$ around $C$. Check that your answers to (b) and (c) are the same. [Hint: If $\mathbf{r}(t)$ is a parametrization of $C$ then the circulation is defined by

$$\begin{aligned}\int_C \mathbf{F} \bullet \mathbf{T}\, ds &= \int \mathbf{F}(\mathbf{r}(t)) \bullet \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \|\mathbf{r}'(t)\|\, dt \\ &= \int \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t)\, dt.\end{aligned}$$

You can take $\mathbf{r}(t) = \langle \cos t, \sin t, \cos t\sin t\rangle$ with $0 \le t \le 2\pi$. At the very end you will need the trig identity $2\cos^2 t = 1 + \cos(2t)$.]

**6. Central Vector Fields are Conservative.** Let $h : \mathbb{R} \to \mathbb{R}$ be any function and consider the vector field $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$$\mathbf{F}(\mathbf{x}) = h(\|\mathbf{x}\|)\frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

(a) Show that $\|\mathbf{F}(\mathbf{x})\|^2 = h(\|\mathbf{x}\|)^2$, so the magnitude of the force only depends on the length of $\mathbf{x}$, not on $\mathbf{x}$ itself. The direction of the force is in the radial direction $\mathbf{x}$. Such vector fields are called *central*.

(b) Consider the scalar field $\ell : \mathbb{R}^3 \to \mathbb{R}$ defined by $\ell(\mathbf{x}) := \|\mathbf{x}\|$. Show that the gradient is

$$\nabla\ell(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

(c) Let $H : \mathbb{R} \to \mathbb{R}$ be any antiderivative of $h$, which exists because any (reasonable) function $\mathbb{R} \to \mathbb{R}$ has an antiderivative. Define the scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ by $f(\mathbf{x}) := H(\ell(\mathbf{x}))$. Use the chain rule to prove that

$$\nabla f(\mathbf{x}) = \mathbf{F}(\mathbf{x}).$$

In other words, any central vector field is conservative.